

[illegible]

Dissimilatory sulfate reduction pathway

The diagram illustrates the dissimilatory sulfate reduction pathway, showing the conversion of sulfate (SO_4^{2-}) to hydrogen sulfide (HS^-) and the subsequent oxidation of HS^- to elemental sulfur ($\text{S}(0)$).

Key components and reactions:

- Sat** (Sulfate Adenylyltransferase) converts SO_4^{2-} to **APS** (Adenosine Phosphosulfate).
- AprBA** (Adenosine Phosphosulfate Reductase) converts APS to SO_3^{2-} (Sulfite).
- DsrBA** (Dissimilatory Sulfurtransferase) and **DsrC** (Dissimilatory Sulfurtransferase) form a cycle where SO_3^{2-} is reduced to HS^- by DsrC, and DsrC is re-oxidized to $\text{S}(0)$ by DsrBA.
- DsrMKJOP** (Dissimilatory Sulfurtransferase) is involved in the final steps of the pathway, including the reduction of MQH_2 to MQ .

Localization:

- Cytoplasm:** Sat, AprBA, and the initial steps of the pathway.
- Membrane:** DsrBA, DsrC, and DsrMKJOP.
- Periplasm:** The final steps of the pathway, including the oxidation of HS^- to $\text{S}(0)$.

Diagram illustrating the dissimilatory sulfide oxidation pathway, showing the flow of sulfur species across the cell envelope (Periplasm, Membrane, Cytoplasm).

Periplasm:

- FccAB:** Cycles between HS^- and HS_n^- via **Sqr**.
- Sqr:** Membrane protein involved in $\text{MQ} \rightarrow \text{MQH}_2$.

Membrane:

- DsrMKJOP:** Membrane protein involved in $\text{MQ} \rightarrow \text{MQH}_2$.
- QmoABC:** Membrane protein involved in $\text{MQ} \rightarrow \text{MQH}_2$.

Cytoplasm:

- DsrC:** Cycles between S^0 and S^{2-} (labeled $\text{S}(0)$).
- DsrBA/DsrL:** Cycles between S^0 and S^{2-} .
- DsrEFH:** Cycles between HS^- and HS_n^- .
- QmoABC:** Cycles between MQ and MQH_2 .

Pathway Flow:

- HS^- is oxidized to HS_n^- by **Sqr**.
- HS_n^- is transported into the cytoplasm.
- HS_n^- is oxidized to S^0 by **DsrMKJOP**.
- S^0 is oxidized to S^{2-} by **DsrC**.
- S^{2-} is oxidized to APS by **AprBA**.
- APS is oxidized to SO_4^{2-} by **Sat**.

Dissimilatory sulfide oxidation pathway

Weissgerber et al. (2014)
Gwak et al. (2022)